

Python Strings & Type Casting — Assignment

Objective

This is a **hands-on, practical assignment** that must be completed using **Google Colab**. The goal is to help you **learn by doing**, not by memorizing concepts. This assignment is designed to help you **deeply understand Python strings and type casting** through structured practice. You will work through **Easy, Medium, and Hard tasks** that gradually build your conceptual clarity and problem-solving skills.

 **Important Rules** -  This assignment must be completed in **Google Colab** -  Each task should be implemented as a **working, practical example** -  **Do NOT write or submit code snippets in the description** (only in your actual solution file) -  Follow task requirements exactly -  Focus on understanding, not memorization -  Write clean, readable solutions in your own words

Easy Tasks (Foundations)

These tasks focus on understanding the **basics of strings and type casting**.

Task 1: String Creation & Types

Requirements: - Create examples of strings using: - Single quotes - Double quotes - Triple quotes (multi-line) - Raw strings - Clearly show the difference in output between **normal strings and raw strings**

Task 2: Escape Sequences

Requirements: - Demonstrate the use of common escape sequences such as: - New line - Tab - Backspace - Quotation marks inside strings - Explain (in comments) what each escape sequence does

Task 3: Basic String Operations

Requirements: - Perform and demonstrate: - String concatenation - Indexing (accessing characters) - Slicing (extracting parts of a string) - Finding the length of a string - Converting strings to upper-case and lower-case - Clearly label the output of each operation

Task 4: Type Casting Basics

Requirements: - Show examples of **implicit type casting** - Show examples of **explicit type casting** using built-in conversion functions - Explain why Python allows or disallows certain conversions

Medium Tasks (Core Understanding)

These tasks focus on **string manipulation, formatting, and comparison**.

Task 5: String Methods

Requirements: - Use and demonstrate the following string methods: - split - join - replace - find - count - Explain the behavior of each method using comments - Highlight how output changes when inputs change

Task 6: String Formatting Techniques

Requirements: - Demonstrate string formatting using: - % operator - str.format() - f-strings - Show how **order matters** in formatting - Clearly explain the difference between old and modern formatting styles

Task 7: String Comparison

Requirements: - Compare strings using comparison operators - Show comparisons with: - Same case strings - Different case strings - Explain how Python uses **Unicode values** for comparison

Task 8: String Repetition

Requirements: - Demonstrate string repetition using the `*` operator - Show repetition with: - Normal strings - Zero repetition - Use repetition to create visual patterns or separators

Hard Tasks (Advanced Concepts)

These tasks focus on **memory behavior, performance concepts, and deep Python internals**.

Task 9: String Interning

Requirements: - Explain what string interning is in your own words - Demonstrate automatic interning - Demonstrate manual interning - Verify memory behavior using identity comparison

Task 10: String Literal Pool

Requirements: - Explain what the string literal pool is - Show how identical string literals share memory - Show cases where strings are NOT pooled

Task 11: Interning vs String Pool

Requirements: - Clearly differentiate between: - String interning (process) - String pool (memory space) - Provide real examples to justify the difference

Task 12: Dynamic vs Static Strings

Requirements: - Demonstrate dynamically created strings - Compare them with static string literals - Explain why dynamic strings are not interned by default

Task 13: Advanced Type Casting

Requirements: - Convert between: - int, float, str, bool - list, tuple, set - dictionary from key-value pairs - complex numbers - Clearly explain which conversions are allowed and which raise errors

Submission Instructions

- Complete this assignment using **Google Colab**
 - Share the **Google Colab notebook link** (access set to *Anyone with the link can view*)
 - Organize your notebook with clear headings for each task
 - Use **markdown cells** to explain concepts and **code cells** to implement them
 - Submit the Colab link on the portal before the deadline
 - Submit your solution as **one Python file**
 - Use **clear comments** to explain your logic
 - Do NOT copy code from slides or internet sources
 - Late submissions may not be accepted
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Evaluation Criteria

- Conceptual clarity
 - Correct use of string operations
 - Proper understanding of type casting
 - Code readability and explanation quality
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 **Note:** This assignment is intentionally designed to test **understanding, not speed**. Focus on learning every concept thoroughly.